2nd Order DEs: Modelling

Questions 1–10

Question 1. The price $p(t)$, in pound sterling, of a particular type of fresh strawberries sold by a large chain of supermarkets is modelled over time t days after a sudden oversupply. Due to delays in adjusting pricing across stores and responding to customer demand, the price is assumed to satisfy

$$4\frac{d^2p}{dt^2} + 4\frac{dp}{dt} + p = 0.$$

At time $t = 0$, the price is £10 per kg and is decreasing at a rate of £2 per day.

a) Find the general solution of the differential equation.
b) Find the particular solution of the differential equation.
c) Determine whether the price, according to the model, ever becomes negative.
d) Find the time at which the rate of decrease of the price is greatest.
e) Explain one limitation of this model in the context of supermarket pricing.

Question 2. A heavy fire door is released and begins to close automatically. Let $\theta(t)$ be the angular displacement, in radians, of the door from the fully closed position at time t seconds. The motion is modelled by

$$\frac{d^2\theta}{dt^2} + 5\frac{d\theta}{dt} + 6\theta = 0.$$

Initially, the door is held open at $\theta = 0.5$ and released from rest.

a) Find the general solution of the differential equation.
b) Find the particular solution of the differential equation.
c) Find the time at which the angular speed of the door is greatest.
d) Find the angular velocity when $t = 1$.
e) Comment on the suitability of this model for large values of t.

Question 3. A carriage on a rotating unfair ride is displaced slightly from its equilibrium position. Let $\theta(t)$ be the angular displacement, in radians, from equilibrium at time t seconds. The motion is modelled by

$$9\frac{d^2\theta}{dt^2} + 6\frac{d\theta}{dt} + 2\theta = 0.$$

At time $t = 0$, $\theta = 0$ and the carriage is given an initial angular velocity of 3 rad s^{-1}.

a) Find the general solution of the differential equation.
b) Find the particular solution of the differential equation.
c) Explain why the motion is oscillatory.
d) Find the maximum angular displacement after $t = 0$.
e) Find the second time at which the carriage comes instantaneously to rest.

Question 4. A water tank is part of a control system that regulates water level. Let $h(t)$, in metres, be the deviation of the water level from its equilibrium level at time t seconds. The system is modelled by

$$\frac{d^2h}{dt^2} + \frac{dh}{dt} + \frac{2}{9}h = 0.$$

At time $t = 0$, the water level is 0.3 m above equilibrium and is falling at a rate of 0.3 m s^{-1}.

a) Find the general solution of the differential equation.
b) Find the particular solution of the differential equation.
c) Find the time at which the water level first reaches equilibrium.
d) Find the greatest deviation of the water level below equilibrium, and the time at which this occurs.
e) Find the rate of change of water level at $t = 3$.

Question 5. A camera on a stabilising tripod is displaced from its equilibrium position, where x m is the horizontal displacement from equilibrium at time t s. The motion is modelled by

$$\frac{d^2x}{dt^2} + a\frac{dx}{dt} + bx = 0,$$

where a and b are positive constants. At time $t = 0$, we have that $x = -0.02$, $\frac{dx}{dt} = 0$ and $\frac{d^2x}{dt^2} = 8$. It is given that the auxiliary equation has repeated roots.

a) Find the values of a and b.
b) Find the particular solution.
c) Describe the motion of the camera as time increases.

Question 6. A car is travelling along a straight road. Let $v(t)$, in m s^{-1}, be its speed at time t seconds. The motion is modelled by

$$\frac{d^2v}{dt^2} + \frac{3}{2}\frac{dv}{dt} + \frac{1}{2}v = 4.$$

Initially, the car has speed 0 m s^{-1} and acceleration 1 m s^{-2}.

a) Find the general solution of the differential equation.
b) Find the particular solution of the differential equation.
c) Find the time at which the acceleration is greatest.
d) Find the limiting speed of the car.
e) Comment on the suitability of this model for large values of t.

Question 7. A room is being heated by a system whose output increases steadily over time. Let $T(t)$, in °C, be the temperature of the room at time t minutes. The temperature is modelled by

$$\frac{d^2T}{dt^2} + 3\frac{dT}{dt} + 2T = 17 + t.$$

At time $t = 0$, the temperature is 8°C and is increasing at 0.25°C per minute.

a) Find the general solution of the differential equation.
b) Find the particular solution of the differential equation.
c) Show that the temperature is always increasing.
d) Find the temperature as $t \to \infty$.
e) Comment on the suitability of this model.

Question 8. A tank is being filled by a pump whose output increases over time, while water simultaneously drains from the tank. Let $h(t)$, in metres, be the height of water above a reference level at time t seconds. The system is modelled by

$$\frac{d^2h}{dt^2} + \frac{dh}{dt} + \frac{1}{4}h = t^2 + 2.$$

At time $t = 0$, it is given that $h = 0$ and $\frac{dh}{dt} = 0$.

a) Find the general solution of the differential equation.
b) Find the particular solution of the differential equation.
c) Find $h''(0)$ and interpret what this means in context.
d) Determine whether, over time, the water level increases without bound.
e) Explain why this model may not be realistic for large values of t.

Question 9. A rotating platform at a fairground is subjected to a periodic driving force from its motor. Let $\theta(t)$, in radians, be the angular displacement from equilibrium at time t seconds. The motion is modelled by

$$80\frac{d^2\theta}{dt^2} + 40\frac{d\theta}{dt} + 25\theta = 136\cos\frac{1}{2}t.$$

At time $t = 0$, the platform is displaced by 0.4 radians from equilibrium and is rotating back towards equilibrium with angular velocity 0.5 rad s^{-1}.

a) Find θ in terms of t.
b) Describe the long-term behaviour of the motion.
c) Write the steady-state motion in the form $R\cos\left(\frac{1}{2}t - \alpha\right)$, where R and α are constants.
d) The platform is regarded as safe only if its steady-state angular displacement never exceeds 6.5 radians in magnitude. Determine whether the platform is safe.

Question 10. A chemical reactor is heated by an external process. The temperature $T(t)$, measured in °C above a reference level, changes over time as heat is supplied to the reactor. A model for the temperature is

$$36\frac{d^2T}{dt^2} + 108\frac{dT}{dt} + 81T = 216e^{-\frac{3}{2}t} + 100e^{\frac{1}{6}t},$$

where t is measured in minutes. Initially, the temperature is 4°C above the reference level and is decreasing at a rate of $\frac{1}{3}$ °C min^{-1}.

a) Find the general solution of the differential equation.
b) Find the particular solution of the differential equation.
c) Describe the behaviour of the temperature as $t \to \infty$.
d) Estimate the temperature in the reactor after 10 minutes to 3 significant figures.
e) Estimate the time at which the temperature in the reactor is 20°C.